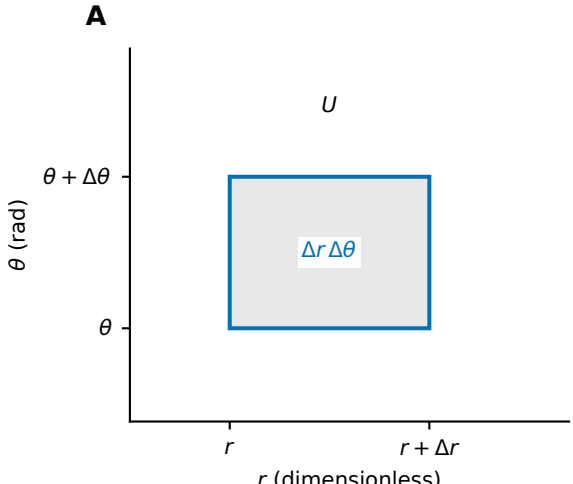
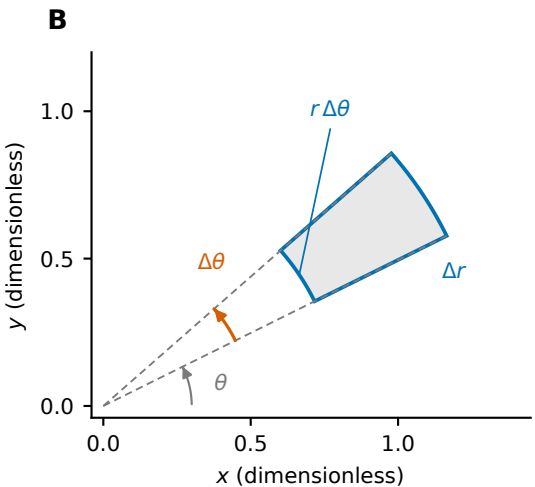


$T(r, \theta) = (r\cos\theta, r\sin\theta)$



$dA = r\,dr\,d\theta$